Branch factor forced by a norm identity

Formal proof packet

27 October 2026

This is a conditional polynomial proof, not a target experiment or cost comparison. The editable diagram, gate CSV, Lean source, and compiler log are beside this PDF.

Checked statement

For polynomials f, u_1, u_2, n, U over a field and a rational point a, assume

$$f(a) = 0, \qquad n^2 - u_1 u_2 U = b^2 f, \qquad u_1(a)u_2(a)U(a) = 0.$$

The checked theorem specializes the norm identity at ordinate zero and uses the polynomial factor theorem to conclude

$$(X - a) \mid n.$$

Norm identity plus rational branch-support root

$\Downarrow$ checked pointwise branch overlap: $n(a) = 0$

$\Downarrow$ checked polynomial factor theorem

$(X - a) \mid n$

Verification and remaining gates

Direct Lean 4.34.1 compilation with Mathlib `d13f23b` exited zero. The theorem axiom audit lists only `propext`, `Classical.choice`, and `Quot.sound`; no `sorryAx`. Lean source SHA-256:

`7e583e29d15de7b4eba06d1cbb93a8eb52812c57039248791389d723317b8051.`

The concrete source-style norm theorem is separately checked. This packet does not show that the compiled runtime numerator satisfies the identity on every branch, that B is a local unit, or that all local orders form the claimed global principal divisor and Pic^0 equation. The frozen 1,260-target rows and matched-rho ratios remain unchanged.